Reg. No. | | | | | | | | | | | [18MTDS201]

SECOND SEMESTER M.TECH EXAMINATION, JUNE 2019

STATISTICAL INFERENCE

Time: 03 Hours **Maximum Marks: 100**

Instructions: **i. Answer FIVE full questions. Question No. 1 and 2 are compulsory.**
ii. Draw neat sketches wherever necessary.
iii. Make suitable assumptions wherever necessary.

1. a. Draw an outline for differences between Machine Learning, Data Science, AI, Deep Learning, and Statistical inferences. **10**

b. Write an introduction to statistical hypothesis and types of errors as well. Give the steps for analyzing the hypothesis testing. **10**

2. a. Explain maximum likelihood estimation with advantages and disadvantages. **08**

b. Let $X_1, X_2, \ldots, X_n$ be a random sample of size n from N (μ σ^2,). Find the maximum likelihood estimations of μ and σ^2 and examine whether they are unbiased and consistent. **12**

3. a. Explain Bayesian Estimation with advantages and disadvantages. **08**

b. A traffic control engineer believes that the cars passing through a particular intersection arrive at a mean rate λ equal to either 3 or 5 for a given time interval. Prior to collecting any data, the engineer believes that it is much more likely that the rate $\lambda = 3$ than $\lambda = 5$. In fact, the engineer believes that the prior probabilities are: **12**

$$P(\lambda=3) = 0.7 \text{ and } P(\lambda=5) = 0.3$$

One day, during a randomly selected time interval, the engineer observes $x = 7$ cars pass through the intersection. In light of the engineer's observation, what is the probability that $\lambda = 3$? And what is the probability that $\lambda = 5$?

Poisson cumulative probability from table for $P(X = 7 \mid \lambda = 3) = 0.022$ and $P(X = 7 \mid \lambda = 5) = 0.105$.

OR

4 a. What are Confidence Intervals and the Central Limit Theorem? **08**

b. A farm has 1000 young pigs with an initial weight of about 50 lbs. They put them on a new diet for 3 weeks and want to know how many pigs to sample so that they can estimate the average weight gain. They want the answer to be within 2 lbs. with 90% confidence. There is no pilot data here. We don't have the time to select out some pigs in order to get an estimate for σ, the standard deviation of the weight gain. **12**

What would be a reasonable measure that would help this farmer to give him some guidance on how to estimate the standard deviation of the weight gain?

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5. a. Explain Rao-Blackwell Theorem with example. **10**

b. Write the properties of the OLS method. **10**

OR

6. a. Explain about Bayesian Hypothesis Testing. **08**

b. Suppose the weights of randomly selected Jain University female students are normally distributed with unknown mean μ and standard deviation σ. A random sample of 10 Jain University female students yielded the following weights (in pounds): **12**

115 122 130 127 149 160 152 138 149 180

Based on the definitions given above, identify the likelihood function and the maximum likelihood estimator of μ, the mean weight of all Jain University female students. Using the given sample, find a maximum likelihood estimate of μ as well.

7. a. Write the procedure for Chi-Square Goodness of Fit Test. **08**

b. Suppose the Jain University student population is 60% female and 40% male. Then, if a sample of 100 students yields 53 females and 47 males, can we conclude that the sample is (random and) representative of the population? Use the chi-square goodness-of-fit statistic to test the hypotheses: **12**

$$H0 : pF = 0.60$$

$$HA : pF \neq 0.60$$

using a significance level of $\alpha = 0.05$.

OR

8. a. Explain Basu's Theorem and its Role in Statistical Inference. **10**

b. Discuss the confidence interval estimation for the parameter estimated through minimum Chi-Square method with example. **10**